## Assignment # 5:

## Definitions.

### Frontend:

The term Frontend refers to the graphical user interface (GUI) that the users can directly interact with, such as navigation menus, design elements, buttons, images, and graphs. It allows your users to interact with your application. In technical terms, a page or screen that the user sees with several UI components is called a Document object Model (DOM).

### Backend:

The backend is the data and infrastructure that make the application work. It stores and processes application data for the users. Sometimes called the server side, the backend of the application manages your web application's overall functionality. When the user interacts with the frontend, the interaction sends a request to the backend in HTTP format. The backend processes the request and returns a response.

### Training and Inferencing:

Artificial intelligence training and artificial intelligence inference are two key elements of the machine learning development lifecycle. Understanding what is Ai training requires knowledge of the fundamentals of machine learning. It refers to the process of teaching an ML model to recognize patterns within a dataset. The training phase, sometimes called the development phase, involves feature engineering, selection and model training.

Inference occurs after the training is complete. During inference, the model is introduced to unseen, real-world scenarios and uses its learning to make accurate predictions.

### Gitignore:

The gitignore file is a simple text file placed in the root directory of your Git repository. It contains patterns that tell Git which files or directories to ignore.

### API:

APIs are mechanisms that enable two software components to communicate with each other using a set of definitions and protocols. For example, the weather bureau's software system contains daily weather data. The weather app on your phone "talks" to this system via APIs and shows you daily weather updates on your phone.

API architecture is usually explained in terms of client and server. The application sending the request is called the client, and the application sending the response is called the server. So in the weather example, the bureau's weather database is the server, and the mobile app is the client.

### API Key:

An API key is a unique identifier used to authenticate software and systems attempting to access other software or systems via an application programming interface, or API.

An API key is a unique series of letters and numbers that's a special ID or secret password for an application or user when making a call to an API.

**Open source Or Local Models:**

The term “open source” refers to the LLM code and underlying architecture being accessible to the public, meaning developers and researchers are free to use, improve or otherwise modify the model.

### Cloud Model:

Cloud service models define the control, management, and abstraction level you have when utilizing cloud services. They determine the type of services available, how much infrastructure you manage, and what level of responsibility the cloud provider assumes.

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Instead of maintaining costly infrastructure and software on-site, you can now use cloud services to meet almost any computing need. These services are delivered through different models, each offering varying levels of control, flexibility, and complexity. Example: Google Cloud, Amazon web service.

### Ollama:

Ollama is a free, open-source platform that lets you download and run large language models (LLMs) directly on your own computer or local server.a

### Tokens:

A token is the basic unit of text an AI model reads and generates – a word, part of a word, or a punctuation mark.

A token can be a whole word ("the"), part of a word ("token" + "ization"), or a single punctuation mark.

### Embedding Models:

An embedding model is a machine learning tool that transforms complex data into simpler, numerical values that machines can understand.

Embedding is a means of representing objects like text, images and audio as numbers in a continuous vector space where the locations of those numbers in space are semantically meaningful to Machine Learning Algorithms.

### Dimensions:

Embedding models convert text into a list of numbers called a vector. The number of values in that vector is called the embedding dimension.

For example, if an embedding model has 384 dimensions, one sentence is represented by a vector containing 384 numbers.

Embedding dimensions mean the number of values (numbers) a model uses to represent a piece of text.